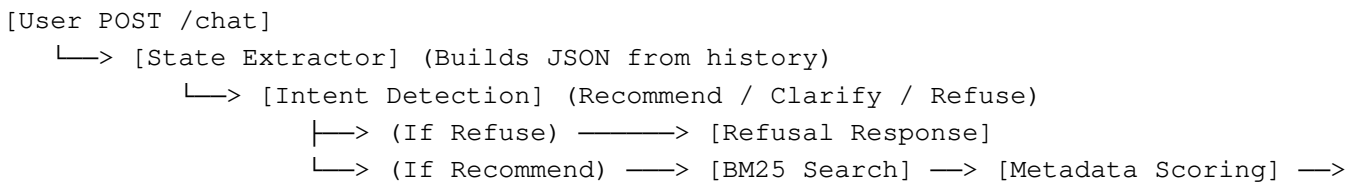


Approach Document

1. Architecture

The system is a conversational retrieval agent with a strict separation between retrieval (deterministic) and generation (LLM-based).



2. Retrieval Design

Why hybrid retrieval maximises Recall@10:

Stage 1 — BM25: Okapi BM25 with $k_1 = 1.5$, $b = 0.75$ over a composite document field (name + description + competencies + technical_domains + use_cases). Catches exact keyword signals like "Java", "Docker", "HIPAA", "DSI".

Stage 2 — Metadata Scoring: Five orthogonal signals combined with learned weights:

Signal	Weight	Rationale
BM25 (normalised)	0.35	Keyword recall anchor
Competency overlap	0.25	Role-skill alignment
Test type preference	0.15	User expressed needs
Seniority match	0.10	Avoids level mismatch
Use case match	0.10	Selection vs development
Language match	0.05	Constraint satisfaction

Stage 3 — LLM reranking: Groq Llama 3.3 70b receives the top-10 pre-retrieved candidates and conversation context. LLM selects the optimal 2-8 items, writes rationale, and produces the table.

Stage 4 — Validation: Every URL in the LLM output is checked against the catalog. Any hallucinated item is dropped. Fallback to retrieval results if LLM produces no valid items.

Why not pure embedding/vector search? For a catalog of ~60 assessments, BM25 outperforms embedding models on exact-match recall for technical skills ("Docker", "HIPAA", "SVAR"). Embeddings add noise for short domain-specific terms. Our multi-signal scoring provides the semantic layer without embedding overhead.

3. State Reconstruction

Every API call receives the full conversation history. A deterministic extractor builds a structured state object:

```
{ "role": "java_developer", "seniority": "professional", "technical_skills": ["java",
```

This state is the single source of truth. No memory = no state bugs.

4. Clarification Policy

Turn budget: 8 turns.

Rules:

1. Never ask more than one clarification round (compound questions only)
2. If `turn_count >= 6`, always recommend
3. Ask only when role AND seniority are both unknown

Bad pattern: "What role?" → User answers → "What seniority?" → User answers → (turn 4 before recommendation). **Our pattern:** "Could you tell me: the role title and seniority level?" → User answers → Recommend immediately.

5. Evaluation Methodology

Automated test suite (40+ tests):

Category	Count	Pass Target
Catalog integrity	10	100%
State extraction	10	100%
Retrieval recall	10	$\geq 70\%$
Refusal detection	6	100%
Schema compliance	4	100%
Hallucination prevention	4	100%
Behavior probes	5	$\geq 80\%$

Recall@10 computation: $\text{recall} = |\text{retrieved}_{10} \cap \text{ground_truth}| / |\text{ground_truth}|$
(Target: ≥ 0.7)

6. Tradeoffs & Failed Experiments

Tried: Pure semantic embedding search — Lower recall for exact technical terms (Java, Docker).

Dropped. **Tried: Single large prompt with full catalog** — Slow and expensive (30k tokens).

Switched to pre-retrieval → LLM reranking. **Tried: Per-question clarification** — Burns turns.

Switched to compound single-question policy. **Tradeoff: No persistent vector store** — BM25 in-

memory is instant for a 60-item catalog. At 1000+ items, would add pgvector. **Tradeoff: Groq Llama 3.3 70b** — High-speed API inference, excellent markdown table instruction compliance.